WG 5

Intelligent Systems

VAGUENESS AND FUZZY LOGIC

Exercise 1: Consider the following statement: "Half of the Dutch men are married". Which of the following statements is true?

A) The author of this statement describes the marital status of Dutch men with a degree of vagueness.
B) The author of this statement describes the marital status of Dutch men with a degree of uncertainty.

Give an example for a statement for the other property?

DEFINITIONS

VAGUENESS
Lack of clarity or precision, that leaves room for interpretation.

UNCERTAINTY
The information provided leaves doubt about its accuracy or completeness. It implies that there is something not fully known within the context of the statement.

Exercise 1: Consider the following statement: "Half of the Dutch men are married". Which of the following statements is true?

A) The author of this statement describes the marital status of Dutch men with a degree of vagueness.
B) The author of this statement describes the marital status of Dutch men with a degree of uncertainty.

When you hear **"Half of the Dutch men are married," it sounds like a specific number, but we don't really know if that's true without seeing the facts.**
It's supposed to be a straight-up half, but the actual percentage might be slightly different. Basically, we're in the dark about whether 'half' is accurate because we don't have the details to confirm it.

Give an example for a statement for the other property?

VAGUENESS: **"Dutch men are very tall”.**

The term "very tall" is subjective and can vary in meaning depending on the person interpreting it.

Exercise 2: The director of our institute wants to reduce the environmental impact of the company by reducing the CO_2 emissions caused by transportation. Therefore, she wants to give a devise a policy to support those who do not live very close (they can come by bicycle) or those who live very far (they should do remote work).

Since you know the director is a big fan of fuzzy set theory, you decide to model this problem with fuzzy sets. How do you model the fuzzy set of "employees who live moderately far away", i.e. those who do not live close, but also not far away?

DEFINITIONS

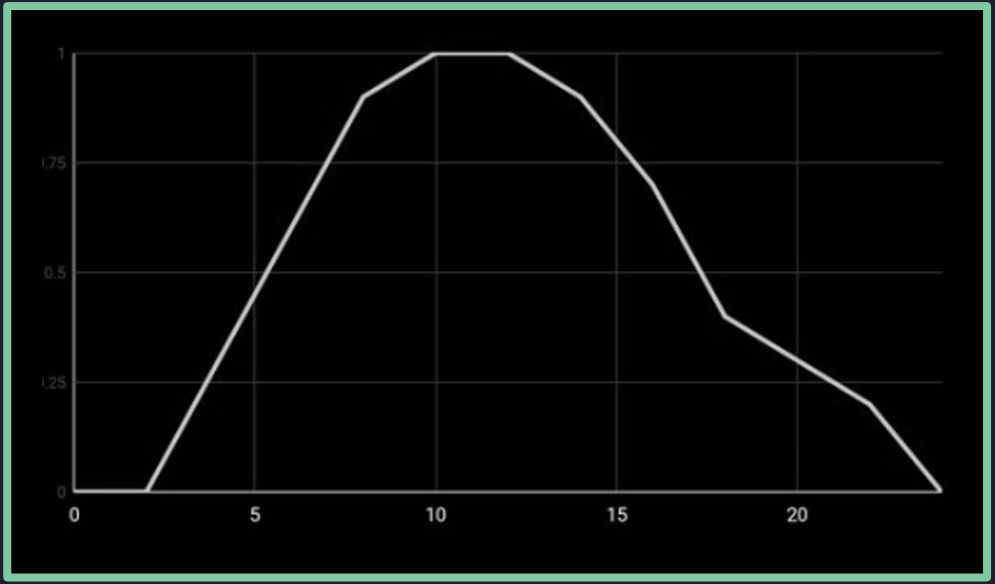
FUZZY SET THEORY

Elements can belong to a set to a certain degree.

MEMBERSHIP FUNCTION

Gives a number between 0 and 1, where 0 means no membership and 1 means full membership

SOLUTION



X - AXIS

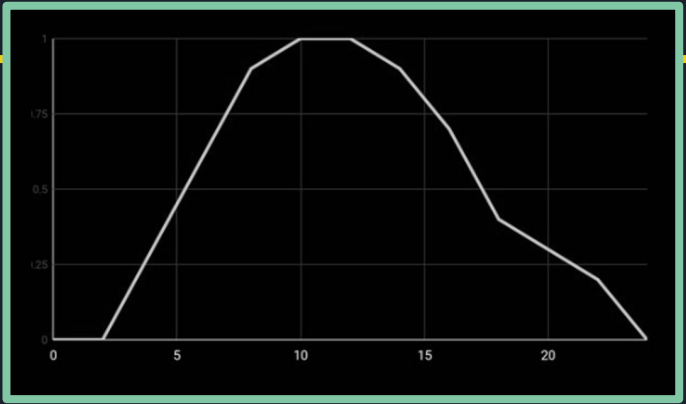
Distance from workplace

Y - AXIS

Degree of belonging to the set of employees who live moderately far away.

As the distance from the workplace increases from 0, the degree to which employees are considered to live moderately far away also increases until it reaches a peak. After this point, as the distance continues to increase, the degree decreases, reflecting that these employees live too far away to be considered as living 'moderately far away' and perhaps should be offered the option to work remotely.

So, if the director wants to use this model, employees who live at distances represented by the peak of the curve would be considered to live at the ideal 'moderately far' distance for the proposed policy.



Exercise 3: Given the following (standard) semantics for Fuzzy Logic for the following operators:

- $f(\neg A) = 1 - f(A)$
- $f(A \vee B) = \max(f(A);f(B))$
- $f(A \& B) = \min(f(A);f(B))$
- $f(A \rightarrow B) = \min(1;1 - f(A) + f(B))$

What is the meaning of $f((P \vee (P \rightarrow Q)) \& (\neg Q \rightarrow \neg P))$ if $f(P)=0.8$ and $f(Q)=0.3$?

DEFINITIONS

In fuzzy logic, variables can have a degree of truth between 0 and 1 (that is, False and True).

The problem provides the truth values for two propositions, P and Q, as well as the standard semantics (rules) for calculating the truth value of negations, conjunctions, disjunctions, and implications.

- f(¬A) = 1 - f(A)
- f(A v B) = max(f(A);f(B))
- f(A & B) = min(f(A);f(B))
- f(A -> B) = min(1;1 – f(A) + f(B))

f((P v (P -> Q)) & (¬Q ->¬P))

f(P)=0.8 and f(Q)=0.3

1) **COMPUTE IMPLICATION (P -> Q)**

f(P -> Q) = min(1; 1 - f(P) + f(Q)) = min(1; 1 - 0.8 + 0.3) = min(1; 0.5) = 0.5

2) **COMPUTE DISJUNCTION (P V (P -> Q))**

f(P V (P -> Q)) = max(f(P); f(P -> Q)) = max(0.8; 0.5) = 0.8

3) **COMPUTE NEGATION -Q**

f(-Q): 1 - f(Q) = 1 - 0.3 = 0.7

4) **COMPUTE NEGATION -P**

f(-P): 1 - f(P) = 1 - 0.8 = 0.2

5) **COMPUTE IMPLICATION: (-Q -> -P)**

f(-Q -> -P) = min(1; 1 - f(-Q) + f(-P)) = min(1; 1 - 0.7 + 0.2) = min(1; 0.5) = 0.5

6) **COMPUTE CONJUNCTION: (P V (P -> Q)) & (-Q -> -P)**

f((P V (P -> Q)) & (-Q -> -P)) = max(f(P V (P -> Q)); f(-Q -> -P)) = max(0.8; 0.5) = 0.8

Exercise 4: Suppose that there are two relations defined as follows, where L5 and L6 stand for Lecture 5 and 6 respectively:

understands

	L5	L6
John	0.4	0.6
Peter	0.3	0.7
Julia	0.1	0.4

contains

	semantics	reasoning
L5	0.8	0.3
L6	0.4	0.8

Fill in the table for the composition of the two relations, i.e. the relation

Understandslecturecontaining = understands o contains

	semantics	reasoning
John		
Peter		
Julia		

The exercise involves two fuzzy relations defined over two sets: the first set is people (John, Peter, Julia), and the second set is lectures(L5 and L6 for Lecture 5 and Lecture 6). The first relation is "understands," which indicates to what degree each person understands each lecture. The second relation is "contains," which indicates how much semantic and reasoning content each lecture contains.

$$f_{R \circ S}(x, z) = \max_y \min(f_R(x, y), f_S(y, z))$$

For each element x from the first set and each element z from the second set, the strength of the relation between x and z in the composed relation is the maximum of the minimum values obtained from all the intermediate elements y.

For each pair (x, z), where x is a person and z is either semantics or reasoning, you look at all possible lectures y (L5 and L6).

You find the minimum value between how well person x understands lecture y and how much lecture y contains of z.

The maximum of these minimums for all lectures y is the value of the composed relation for the pair (x, z).

Calculate the composed relation value for (John, semantics)

- Calculate the minimum of John's understanding of L5 and L5's semantic content: min(0.4, 0.8) = 0.4.
- Do the same for L6: min(0.6, 0.4) = 0.4.
- Take the maximum of these minimums: max(0.4, 0.4) = 0.4.

understands

	L5	L6
John	0.4	0.6
Peter	0.3	0.7
Julia	0.1	0.4

contains

	semantics	reasoning
L5	0.8	0.3
L6	0.4	0.8

Fill in the table for the composition of the two relations, i.e. the relation

Understandslecturecontaining = understands o contains

	semantics	reasoning
John		
Peter		
Julia		

	semantics	reasoning
John	0.4	0.6
Peter	0.4	0.7
Julia	0.4	0.4

```
John, reasoning = max(min(0.4,0.3),min(0.6,0.8)=0.6

Peter, semantics = max(min(0.3,0.8),(0.7,0.4)= 0.4

Peter, reasoning = max(min(0.3,0.3),(0.7,0.8))=0.7

Julia, semantics  = max(min(0.1,0.8),min(0.4,0.4))=0.4

Julia, reasoning = max(min(0.1,0.3),min(0.4,0.8) = 0.4
```

understands

	L5	L6
John	0.4	0.6
Peter	0.3	0.7
Julia	0.1	0.4

contains

	semantics	reasoning
L5	0.8	0.3
L6	0.4	0.8

Fill in the table for the composition of the two relations, i.e. the relation

Understandslecturecontaining = understands o contains

	semantics	reasoning
John		
Peter		
Julia		

Exercise 4: Consider the full joint probability distribution P(A,B,C) shown in the table to the right:

a) What is P(A = true)?

b) What is P(B ^ C = true)?

A	B	C	P(A, B, C)
false	false	false	0.05
false	false	true	0.10
false	true	false	0.03
false	true	true	0.25
true	false	false	0.15
true	false	true	0.02
true	true	false	0.07
true	true	true	0.33

DEFINITION

Full Joint Probability Distribution

A table that lists the probability of every possible combination of these events.

Each row in the table shows a unique scenario. For example, one row might show the probability of A happening, but not B or C. Another row shows the probability of A and B happening, but not C, and so on.

a) What is P(A = true)?

To find the probability of event A occurring (regardless of what happens with B and C), you look at all the rows where A is true and add up those probabilities. It's like saying, "What are all the ways A can happen, and how likely are they in total?" You're summing up the chances for all scenarios where A is true.

P(A = true) = 0.15 + 0.02 + 0.07 + 0.33 = 0.57

A	B	C	P(A, B, C)
false	false	false	0.05
false	false	true	0.10
false	true	false	0.03
false	true	true	0.25
true	false	false	0.15
true	false	true	0.02
true	true	false	0.07
true	true	true	0.33

b) What is P(B ^ C = true)?

To find the probability of both B and C occurring simultaneously, you focus on rows where both B and C are true, regardless of A's status. It's like asking, "In what scenarios do both B and C happen, and how likely are those scenarios in total?" Here, you're adding up the probabilities for the specific scenario where B and C are both true

P(B & C= true) = 0.25 + 0.33 = 0.58

A	B	C	P(A, B, C)
false	false	false	0.05
false	false	true	0.10
false	true	false	0.03
false	true	true	0.25
true	false	false	0.15
true	false	true	0.02
true	true	false	0.07
true	true	true	0.33

Exercise 5:

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

Given the full joint distribution as above, calculate the following:

a. P(toothache).
b. **P**(Cavity).
c. **P**(Toothache|cavity).
d. **P**(Cavity|(toothache v catch)).

Marginal Probability

This is the probability of an event occurring, regardless of the other variables. In our exercise, if you want to know the chance of a "toothache" without caring whether there's a "cavity" or whether the "catch" test is positive, you would look at the marginal probability. You calculate it by adding up all the probabilities of having a toothache across the different scenarios for cavities and catch. It's like asking, "What's the chance of it raining today?" without worrying about factors like temperature or wind.

Vector Probability

It refers to the set of probabilities for all possible outcomes of a particular variable, which can be represented as a vector. For instance, the vector of probabilities for "Cavity" would include the probability of having a cavity (true) and not having a cavity (false), expressed as (0.2, 0.8). It's like saying, "Here are the chances for each thing that can happen with our cavity situation: either we have one or we don't."

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

a. P(toothache).
b. **P**(Cavity).
c. **P**(Toothache|cavity).
d. **P**(Cavity|(toothache v catch)).

a. This asks for the probability that Toothache is true.
P(toothache) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2

b. This asks for the vector of probability values for the random variable Cavity.
It has two values, which we list in the order ⟨true , false⟩.
First add up 0.108 + 0.012 + 0.072 + 0.008 = 0.2.
Then, add up 0.016 + 0.064 + 0.144 + 0.576 = 0.8.
Then we have P(Cavity) = ⟨0.2, 0.8⟩.

c. This asks for the vector of probability values for Toothache, given that cavity is true. Do not forget to normalize the values, the sum should be one.
P(A|B) = P(A,B) / P(B)
P(Toothache |cavity) = ⟨ (.108 + .012)/0.2, (0.072 + 0.008)/0.2⟩= ⟨0.6, 0.4⟩

d. This asks for the vector of probability values for Cavity, given that either toothache or catch is true.
P(toothache v catch) = 0.108+0.012+0.016+0.064+ 0.072 + 0.144 = 0.416.

P(Cavity|toothache v catch) = ⟨(0.108 + 0.012 + 0.072)/0.416, (0.016 + 0.064 + 0.144)/0.416⟩= ⟨0.4615, 0.5384⟩

Exercise 6:

You throw two fair dice, one green and one red, and observe the numbers uppermost. Decide whether the following pairs of events are (in)dependent.

A: the sum is even

B: the red die is even

DEFINITIONS

Events Defined:

- Event A: The sum of the numbers on the two dice is even.
- Event B: The number on the red die is even.

Independence of Events:

Two events, A and B, are independent if the occurrence of one does not affect the occurrence of the other. Mathematically, A and B are independent if $P(A \cap B) = P(A) \times P(B)$, where $P(A \cap B)$ is the probability of both A and B occurring together, $P(A)$ is the probability of A occurring, and $P(B)$ is the probability of B occurring.

A: the sum is even

List Outcomes: There are 6 possible outcomes for each die. When you roll two dice, you can create a 6x6 grid (or matrix) of possible outcomes. Each cell in the grid represents a unique pair of rolls—one from the first die (green) and one from the second die (red).

Determine Even Sums: An even sum can occur in one of the following ways:

- Even + Even: Both dice show an even number. Since there are three even numbers on a die (2, 4, 6), there are $3 \times 3 = 9$ combinations here.
- Odd + Odd: Both dice show an odd number. Similarly, since there are three odd numbers on a die (1, 3, 5), there are $3 \times 3 = 9$ combinations here.

Total Even Sums: Add the number of even+even and odd+odd combinations:

9 (even+even) +9 (odd+odd) =18

ANSWER: 18/36 = 1/2

B: the red die is even

The probability of the red die being even (2, 4, or 6) is also **1/2**, because three out of the six possible outcomes are even.

ARE THE EVENTS A AND B INDEPENDENT?

$$P(A|B) = P(A).$$

Determine Even Sums: An even sum can occur in one of the following ways:

- Even + Even: Both dice show an even number. Since there are three even numbers on a die (2, 4, 6), there are 3×3=9 combinations here.
- Odd + Odd: Both dice show an odd number. Similarly, since there are three odd numbers on a die (1, 3, 5), there are 3×3=9 combinations here. Total combinations: 9 + 9 = 18

P(A|B) = (P(B|A)*P(A))/P(B) = (9/18*1/2)/1/2 = 1/2 = P(A)

Then, A and B are independent.

Exercise 8:

Given the following information:

- The probability that someone with a cold also has fever is 80%.
- The prior probability of having a cold is 0.0001.
- More or less 1 out of 1000 people in the population has fever.

I have fever. Calculate with Bayes' Rule the probability that I have cold.

$$P(A|B) = (P(B|A) \cdot P(A)) / P(B)$$

Where:

- $P(A|B)$ is the conditional probability of event A occurring given that B is true.
- $P(B|A)$ is the conditional probability of event B occurring given that A is true.
- $P(A)$ is the probability of event A.
- $P(B)$ is the probability of event B.

The posterior probability of A given B is equal to the likelihood of B given A times the prior probability of A, all divided by the probability of B.

SOLUTION

- The probability that someone with a cold also has fever is 80%.
- The prior probability of having a cold is 0.0001.
- More or less 1 out of 1000 people in the population has fever.

- P(fever|cold) = 0.8
- P(cold) = 0.0001
- P(fever) = 0.001
- P(cold|fever) = (P(fever|cold) x P(cold)) / P(fever) = (0.8 x 0.0001) / 0.001 = 0.08
- The probability that I have a cold given I have fever is 8%

Exercise 9

B	P(B)
+b	0.001
-b	0.999

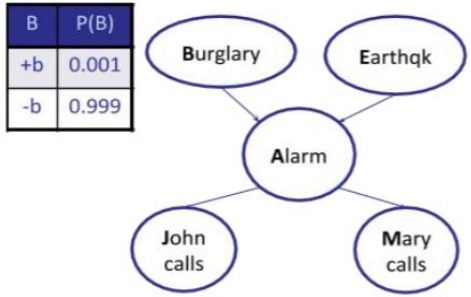
E	P(E)
+e	0.002
-e	0.998

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

Exercise 9



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+e	0.002
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B	E	A	P(A|B,E)
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+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

WHAT IS THE PROBABILITY THAT AN EARTHQUAKE HAPPENS?

Earthquake is independent of other probabilities, so it comes directly from the table, +e = 0.002.

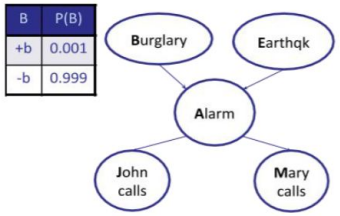
WHAT IS THE PROBABILITY THAT THE ALARM RINGS?

The alarm is dependent on whether a burglary and/or a burglary is happening.
The probability that the alarm rings, when there is, or there is no, earthquake and there is, or there is no, burglary is:

P(a+, b+/-,e+/-) =
P(a+, b+, e+) + P(a+, b+, e-) + P(a+, b-, e+) + P(a+, b-, e-)

P(a+, b+/-,e+/-) =
P(a+, b+, e+) + P(a+, b+, e-) + P(a+, b-, e+) + P(a+, b-, e-)

Exercise 9



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E	P(E)
+e	0.002
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-a	+j	0.05
-a	-j	0.95

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+a	+m	0.7
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B	E	A	P(A|B,E)
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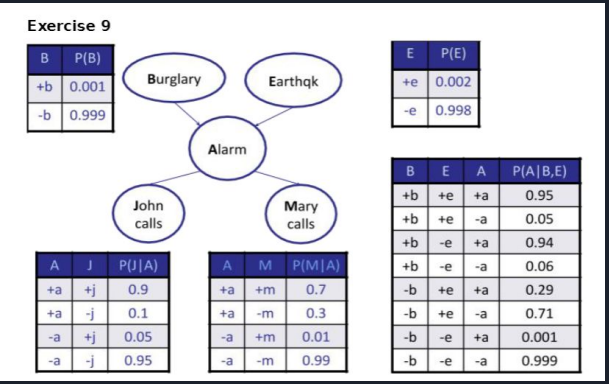
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

P(b+, e+) * P(a+|b+,e+) = 0.002*0.001*0.95 = 0,0000019
+
P(b+, e-) * P(a+ | b+, e-) = 0.001 * 0.998 * 0.94 = 0.00093812
+
P(b-, e+) * P(a+ | b-, e+) = 0.999 * 0.002 * 0.29 = 0.00057942
+
P(b-, e-) * P(a+ | b-, e-) = 0.999 * 0.998 * 0.001 = 0.000997002
= 0,002516442 = P(a+)

WHAT IS THE PROBABILITY THAT JOHN CALLS?

This probability is conditional to whether the alarm rings or n
(look at the table).
But we're lucky since we just computed the probability that the
alarm rings in the previous point!



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+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
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-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

P(a+) = 0,002516442
P(a-) = 1 - P(a+)= 1 - 0,002516442

P(j+, a+/-) =

P(j+,a+) + P(j+,a-) =

P(j+,a+) = P(a+)* p(j+|a+) = 0,002516442 * 0.9 = 0,0022647978

+

P(j+,a-) = P(a-)*p(j+|a-) = (1-0,002516442) * 0.05 = 0,0498741779

= 0.0521389757